

- a What is the greatest value of y ?
- b For what value of x is $3x - x^2$ a maximum?
- c Find the values of x when $y = -3$.
- d For what range of values of x is $3x - x^2$ positive?

- 5 Table 7.6 is a table of values for $y = x^2 + 3x - 1$ from $x = -4$ to $x = 1$.

x	-4	-3	-2	-1	0	1
y	3	-1	-3	-3	-1	3

Table 7.6

- a Use a scale of 2 cm to 1 unit on both axes to draw the graph of $y = x^2 + 3x - 1$.

Use the graph to find

- b the minimum value of y ,
- c the value of y when $x = -3.5$,
- d the values of x when $y = 3$.

- 6 Table 7.7 is a table of values of $y = x - x^2$ from $x = -2$ to $x = 3$.

x	-2	-1	0	1	2	3
y	-6	-2	0	0	-2	-6

Table 7.7

Choose a suitable scale and draw the graph of $y = x - x^2$. Use the graph to

- a state the range of values of x for which y is positive,
- b find the value of x for which $x - x^2$ is a maximum,
- c find the maximum value of y ,
- d find the value of y when $x = 2.2$,
- e find the values of x when $y = -5$.

- 7 a Given that $y = x^2 - 2x - 1$, copy and complete Table 7.8.

x	-2	-1	0	1	2	3	4
x^2	4	1	0	1	4		
$-2x$	4	2	0	-2	-4		
-1	-1	-1	-1	-1	-1	-1	-1
y	7	2		-2			

Table 7.8

- b Use a scale of 2 cm to 1 unit on both axes to draw the graph of $y = x^2 - 2x - 1$.

Use your graph to find

- c the value of y when $x = 1.3$,
- d the values of x when $y = 6$,
- e the minimum value of y .

- 8 a Given that $y = 4 + 3x - x^2$, copy and complete Table 7.9.

x	-1	0	1	2	3	4
4	4	4	4	4	4	4
$+3x$	-3		3		9	
$-x^2$	-1		-1		-9	-1
y	0		6		4	

Table 7.9

- b Using a suitable scale, draw the graph of $y = 4 + 3x - x^2$.

Use your graph to

- c state the range of values of x for which $4 + 3x - x^2$ is positive,
- d find the value of y when $x = -0.6$,
- e find the values of x when $y = 3$,
- f find the maximum value of y .

- 9 Using a scale of 2 cm to 1 unit on both axes, draw the graph of $y = x^2$ for values of x from -3 to $+3$. Read off the values of y when

$x =$ a 1.5, b -0.7,
c 2.3, d -1.3.

Find the values of x when

$y =$ e 5, f 8,
g 1.9, h 0.8.

- 10 a Draw the graph of the function $x^2 - 3x$ from $x = -1$ to $x = +4$.
- b Compare your graph with that of Fig. 7.6. What is the main difference between the two graphs?
 - c For what range of values of x is $x^2 - 3x$ negative?
 - d What is the minimum value of $x^2 - 3x$?

- 11 a Draw the graph of $y = x^2 - 2x$ from $x = -2$ to $x = +4$.
- b Compare the graph with that of question 7. What is the main difference between the two graphs?
- c What is the minimum value of y ?
- d Find the values of x when $y = 7$.

- 12 a Draw the graph of $y = 5 + 4x - x^2$ from $x = -1$ to $x = +5$.

Use your graph to find

- b the value of y when $x = 1.7$,
- c the values of x when $y = 4$,
- d the range of values of x for which $y > 7$.

SUMMARY

- 1 A *linear function* of x is one that contains terms in x of power 1 only. The graph of a linear function is always a *straight line*.
- 2 A *quadratic function* of x is one in which 2 is the highest power of x . The graph of a quadratic function is a curve known as a *parabola*.
- 3 When drawing graphs, it is important to choose x - and y -axes scales carefully and to position the *origin* in a suitable place. The *range of values* of x and y will help to determine the scales and a *table of values* assists in plotting points.
- 4 When drawing a parabola, it is usually helpful to position your hand inside the curve, drawing from left to right if right-handed (or from right to left if left-handed).
- 5 When the coefficient of x^2 is positive, the parabola is U-shaped and it has a *minimum point*; when the coefficient of x^2 is negative, the parabola is n-shaped and it has a *maximum point*.

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Recall and use set notation and language
- 2 Identify the complement of a set
- 3 Write and interpret sets of values using set-builder notation
- 4 Use universal sets, complements and disjoint sets to solve practical problems
- 5 Use Venn diagrams, with up to three intersecting sets, to solve practical problems.

Teaching and learning materials

Teacher: Loops of string to make Venn diagrams; charts showing Venn diagrams.

8.1 Sets (quick revision)

Example 1

Given $\mathcal{U} = \{f, o, r, m, u, l, a, e\}$, $A = \{f, r, a, m, e\}$ and $B = \{r, u, l, e\}$,

- draw a Venn diagram showing A , B and $\mathcal{U}$;
- list the elements of $A \cap B$;
- find $n(A \cup B)$.

a Fig. 8.1 is the required Venn diagram.

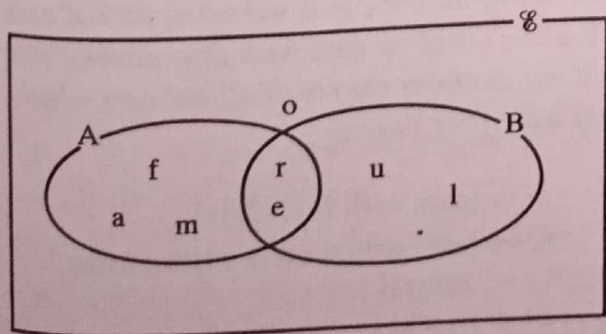


Fig. 8.1

- $A \cap B$ is the intersection of sets A and B , i.e. the set whose elements are members of both A and B .

$$A \cap B = \{r, e\}$$

- $A \cup B$ is the union of sets A and B , i.e. the set whose elements are members of A or B or both A and B .

$$A \cup B = \{f, r, a, m, e, u, l\}$$

Notice that although r and e are elements of both A and B , there is no need to write them down twice.

$n(A \cup B)$ is the number of elements in the set $A \cup B$.

$$n(A \cup B) = 7$$

Table 8.1 contains some of the symbols and language of sets that appeared in Chapter 5.

symbols	meaning
$P = \{a, b, c, d\}$	P is the set a, b, c, d
$Q = \{1, 2, 3, \dots\}$	$\dots$ means 'and so on'
$\emptyset$ or $\{\}$	the empty set
$\mathcal{U}$ or $\mathcal{U}$	the universal set
$\in$	is a member of
$\notin$	is <i>not</i> a member of
$\subset$	is a subset of
$\not\subset$	is <i>not</i> a subset of
$\supset$	contains
$P \cup Q$	union of P and Q
$P \cap Q$	intersection of P and Q
$n(S)$	number of elements in S

Table 8.1

Exercise 8a (Revision)

- 1 Make each of the following true by writing either $\in$ or $\notin$ in place of the $*$.
 - $9 * \{2, 4, 6, 8, 10\}$
 - $15 * \{3, 6, 9, \dots, 24\}$
 - $44 * \{5, 10, 15, 20, \dots\}$
 - $R * \{a, b, c, d, \dots, z\}$

② Find $n(X)$ when $X =$

- a** {h, o, u, s, e} **b** {toes on your feet}
c {0, 1, 2} **d** {months in a year}
e {5, 5, 6, 6} **f** {11, 12, 13, ..., 22}

③ Give three examples of an empty set.

④ Give three examples of an infinite set.

⑤ Write down all the subsets of the following.

- a** {3, 4, 5} **b** {x, y}
c {0, 2} **d** {f, o, u, r}

⑥ Write down the following using symbols.

- a** 2 and 6 are a subset of the factors of 18
b {trees} is not a subset of {metal objects}
c {vehicles} contains {buses}
d children are members of the human race

⑦ If $\mathcal{E} = \{\text{months of the year}\}$, $F = \{\text{first eight months of the year}\}$, $Y = \{\text{months ending in y}\}$, draw a Venn diagram to show the relationship between F , Y and $\mathcal{E}$.

⑧ Which of the following pairs of sets are disjoint? If the sets are not disjoint, write down two members of the intersection.

- a** {prime factors of 24}, {prime factors of 55}
b {multiples of 5}, {multiples of 7}
c {Zimbabwe, Zaria, Zambia}, {countries of Africa}
d {letters of *bull*}, {letters of *cow*}

⑨ If $\mathcal{U} = \{2, 4, 6, 8, \dots, 20\}$

$M = \{\text{multiples of 6 less than 20}\}$

$L = \{\text{even numbers less than 14}\}$

write down the members of the following sets.

- a** $M \cup L$ **b** $M \cap L$ **c** $M \cup \mathcal{U}$
d $\mathcal{U} \cap M$ **e** $\mathcal{U} \cap L$ **f** $L \cup \mathcal{U}$

⑩ **a** Draw a Venn diagram to represent the data of question 9.

b Hence find

- i** $n(M \cup L)$, **ii** $n(M \cap L)$.

8.2 Complement of a set

In Fig. 8.2, A is a subset of the universal set $\mathcal{E}$, i.e. $A \subset \mathcal{E}$.

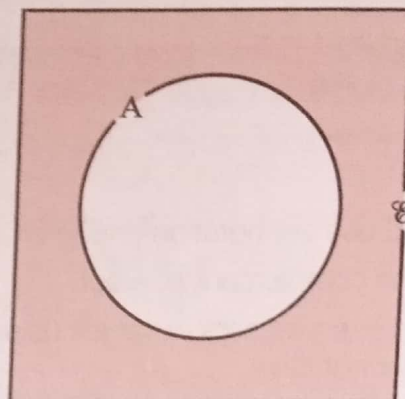


Fig. 8.2

The **complement** of A is the set which contains all those elements of $\mathcal{E}$ which are *not* members of A . A' is short for the complement of A . In Fig. 8.2 the shaded region represents A' .

Example 2

If $\mathcal{E} = \{1, 2, 3, 4, 5\}$, $A = \{1, 3\}$ and $B = \{3, 4\}$ find

a A' , **b** B' , **c** $(A \cap B)'$, **d** $(A \cup B)'$.

a $A' = \{2, 4, 5\}$

b $B' = \{1, 2, 5\}$

c $A \cap B = \{3\}$
 $(A \cap B)' = \{1, 2, 4, 5\}$

d $A \cup B = \{1, 3, 4\}$
 $(A \cup B)' = \{2, 5\}$

Example 3

$\mathcal{E}$ is the set of all cars, R is the set of all red cars and F is the set of all cars with four doors. Show on a Venn diagram the set of all red cars which do not have four doors.

$F = \{\text{cars with four doors}\}$

$F' = \{\text{cars which do not have four doors}\}$

$R \cap F' = \{\text{red cars which do not have four doors}\}$

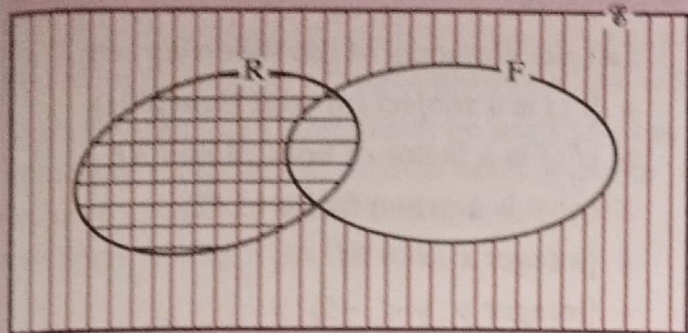


Fig. 8.3

In Fig. 8.3, the region with horizontal shading represents R . The region with vertical shading represents F . The region with cross-shading represents $R \cap F$, the set of all red cars which do not have four doors.

Exercise 8b

- ① Given $\mathcal{U} = \{1, 2, 3, 4, 6, 8\}$, write down the complements of the following sets.

a $\{1, 2, 3\}$	b $\{4, 6, 8\}$
c $\{2, 4\}$	d $\{1, 3, 6, 8\}$
e $\{6\}$	f $\{1, 2, 3, 4, 6\}$
g $\emptyset$	h $\mathcal{U}$
i $\{1, 2, 3, 6\} \cup \{1, 2, 8\}$	
j $\{1, 2, 3, 6\} \cap \{1, 2, 8\}$	
- ② If $\mathcal{U} = \{a, b, c, d, e\}$, $A = \{a, c, e\}$ and $B = \{b, e\}$, list the members of the following sets.

a A'	b B'	c $A' \cup B$
d $A \cup B'$	e $A' \cap B$	f $A \cap B'$
g $A' \cup A$	h $B' \cap B$	i $(A \cup B)'$
j $A' \cup B'$	k $(A \cap B)'$	l $A' \cap B'$
- ③ Compare the answers to parts i and l of question 2. What do you notice?
- ④ Compare the answers to parts j and k of question 2. What do you notice?
- ⑤ If $\mathcal{U} = \{1, 2, 3, 4, 5, \dots, 10\}$, $A = \{1, 2, 5, 7\}$, $B = \{1, 3, 6, 7\}$, write down the sets A' , B' , $(A \cap B)'$ and $(A \cup B)'$.
- ⑥ Using the sets of question 5, show that $A' \cup B' = (A \cap B)'$ and $A' \cap B' = (A \cup B)'$.
- ⑦ Use Venn diagrams to illustrate the results of question 6.

- ⑧ If $\mathcal{U} = \{\text{days of the week}\}$, $S = \{\text{days which contain the letter s}\}$ and $N = \{\text{days which contain six letters}\}$
 - a list the members of the sets S , N , S' , N' ,
 - b list the members of the set
 - i $(S \cup N)'$, ii $(S \cap N)'$,
 - c hence, without further working, list the members of the sets
 - i $S' \cap N'$, ii $S' \cup N'$.
- ⑨ Use a Venn diagram to show that if $A \subset B$ then $B' \subset A'$, and vice versa.
- ⑩ If $\mathcal{U} = \{\text{all teachers}\}$, $M = \{\text{mathematics teachers}\}$ and $W = \{\text{teachers who are women}\}$, show on a Venn diagram the set of all mathematics teachers who are men.

8.3 Set-builder notation

$A = \{\text{integer } x : x < 4\}$ is short for 'A is the set of integers, x , such that x is less than 4'. This is an example of **set-builder notation**. In set-builder notation, the colon ($:$) stands for 'such that'. If set-builder notation is expressed properly, it is always possible to list the elements of the given set. For example, in this case

$$A = \{\dots, -2, -1, 0, 1, 2, 3\}$$

Notice that the elements of A are restricted to integral or whole-number values. The letter $\mathbb{Z}$, sometimes written $\mathbb{Z}$, is often reserved for the set of integers:

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, +1, +2, +3, \dots\}$$

Hence set A can be rewritten

$$A = \{x : x < 4, x \in \mathbb{Z}\}$$

i.e. 'A is the set of values of x such that x is less than 4, where x is an integer'.

If, however, the elements of A are restricted to natural numbers, i.e. positive whole-numbers, then

$$A = \{1, 2, 3\}$$

The letter $\mathbb{N}$, sometimes written $\mathbb{N}$, is often reserved for the set of natural numbers:

$$\mathbb{N} = \{1, 2, 3, \dots\} \text{ or } \mathbb{N} = \{1, 2, 3, \dots\}$$

Hence set A can be rewritten:

$$A = \{x : x < 4, x \in \mathbb{N}\}$$

i.e. 'A is the set of values of x such that x is less than 4, where x is a natural number'.

Example 4

List the members of the set $\{x : x \geq -2, x \in \mathbb{Z}\}$.

$$\{x : x \geq -2, x \in \mathbb{Z}\} = \{-2, -1, 0, 1, \dots\}$$

Remember the meanings of the following symbols:

$>$ is greater than

$\geq$ is greater than or equal to

$<$ is less than

$\leq$ is less than or equal to

Example 5

Given that x is an integer, list the members of $\{x : x < 4\} \cap \{x : x \geq -2\}$.

If $x \in \mathbb{Z}$,

$$\{x : x < 4\} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$$

$$\{x : x \geq -2\} = \{-2, -1, 0, 1, 2, 3, 4, \dots\}$$

$$\{x : x < 4\} \cap \{x : x \geq -2\} = \{-2, -1, 0, 1, 2, 3\}$$

Notice, in Example 5, that the expression $\{x : x < 4\} \cap \{x : x \geq -2\}$ can be shortened to $\{x : -2 \leq x < 4\}$, i.e. 'the set of values, x , such that x is greater than or equal to -2 and less than 4 '. The expression $-2 \leq x < 4$ is a **range of values** of x .

Example 6

List the members of the set $\{y : 1 < y \leq 6, y \in \mathbb{N}\}$.

$$\{y : 1 < y \leq 6, y \in \mathbb{N}\} = \{2, 3, 4, 5, 6\}$$

Example 7

If $A = \{1, 2, 3, 4\}$, rewrite A using set-builder notation.

There are many possible solutions. For example

$$\text{Either } A = \{x : 1 \leq x \leq 4, x \in \mathbb{N}\}$$

$$\text{or } A = \{x : 0 < x < 5, x \in \mathbb{Z}\}$$

$$\text{or } A = \{x : x < 5, x \in \mathbb{N}\}$$

The letter $\mathbb{Q}$, sometimes written $\mathbb{Q}$, is often reserved for the set of rational numbers.

$$\mathbb{Q} = \{\text{numbers of the form } \frac{a}{b} \\ \text{where } a \in \mathbb{Z} \text{ and } b \in \mathbb{N}\}$$

$$\mathbb{Q} = \{\frac{a}{b} : a \in \mathbb{Z}, b \in \mathbb{N}\}$$

Exercise 8c

- ① List the elements of the following sets.
 - a $\{t : t \text{ is a subject on your timetable}\}$
 - b $\{f : f \text{ is a factor of both 24 and 30}\}$
 - c $\{p : p \text{ is a prime factor of 42}\}$
 - d $\{\text{integer } x : x > 8\}$
 - e $\{\text{integer } y : y < -5\}$
 - f $\{x : 3 < x < 5, x \in \mathbb{N}\}$
 - g $\{x : 3 < x \leq 5, x \in \mathbb{N}\}$
 - h $\{x : -2 < x < -1, x \in \mathbb{Z}\}$
- ② If $\mathbb{E} = \{\text{positive integers}\}$, list the members of the following sets.
Note: 0 (zero) is neither positive nor negative.
 - a $\{x : x > 7\}$
 - b $\{x : x > -3\}$
 - c $\{x : x < -1\}$
 - d $\{x : x < 10\}$
 - e $\{x : x - 3 = 0\}$
 - f $\{x : x + 3 = 9\}$
 - g $\{x : x + 2 = 0\}$
 - h $\{x : 2x = 7\}$
- ③ Assuming that x is an integer, state the lowest and highest values of x for each of the following ranges.
 - a $-2 < x < 5$
 - b $-8 \leq x \leq 9$
 - c $-5 < x \leq 0$
 - d $2 \leq x < 10$
 - e $4 < x < 6$
 - f $-8 < x < -1$
 - g $-7 \leq x < 1$
 - h $-5\frac{2}{3} < x < 1\frac{3}{4}$
- ④ Using the inequality symbol $<$, write the following in set-builder notation.
 - a $\{0, 1, 2, 3\}$
 - b $\{-1, 0, 1\}$
 - c $\{-3, -2, -1, 0\}$
 - d $\{-9, -8, \dots, 3, 4\}$
 - e $\{9, 10\}$
 - f $\{-8, -7, \dots, -1\}$
- ⑤ If $x \in \mathcal{U}$ and $\mathcal{U} = \{1, 2, 3, \dots, 10\}$, list the members of the following subsets of $\mathcal{U}$.
 - a $\{x : x \text{ is an odd number}\}$
 - b $\{x : x < 7\}$
 - c $\{x : x + 2 \geq 7\}$
 - d $\{x : x - 2 = 2\}$
 - e $\{x : 3x - 2 = 19\}$
 - f $\{x : x \text{ is a factor of 24}\}$

8.4 Venn diagrams and problem solving

Venn diagrams can sometimes be used to store numerical information. In such cases it is also possible to use the Venn diagrams to solve problems arising from the data.

Example 8

50 students were asked what they did last night. 16 said they read a book, 41 said they watched television. If 7 said they did neither, how many did both?

Let $\mathcal{E} = \{\text{all students}\}$, $B = \{\text{book readers}\}$ and $T = \{\text{television watchers}\}$. It is required to find $n(B \cap T)$. Let $n(B \cap T) = x$.

Fig. 8.4 is a Venn diagram containing the given information. The numbers in the regions of the Venn diagram represent the numbers of elements in the regions.

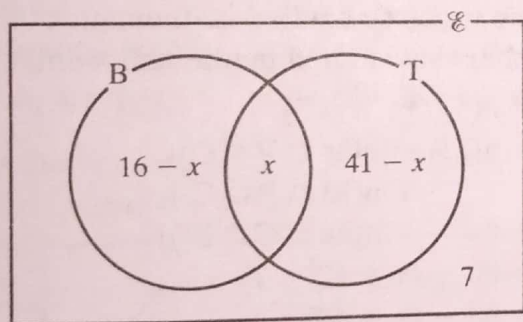


Fig. 8.4

$$\begin{aligned}\text{Since } n(T) &= 41, n(B' \cap T) \\ &= 41 - x.\end{aligned}$$

$$\begin{aligned}\text{Similarly, } n(B \cap T') &= 16 - x \\ \text{and } n(B \cup T)' &= 7\end{aligned}$$

The totals for the regions must add up to the number of people in the universal set:

$$\begin{aligned}x + (16 - x) + (41 - x) + 7 &= 50 \\ \Leftrightarrow 64 - x &= 50 \\ \Leftrightarrow x &= 14\end{aligned}$$

14 students read a book *and* watched television.

Note: The 16 who read a book includes the 14 who also watched television.

Example 9

In a survey of students, it was discovered that 76 like history, 98 like geography and 87 like mathematics. 15 like all the three subjects, 39 like history and geography, 37 like history and mathematics, 48 like geography and mathematics and 45 like none of the subjects.

- Show this information using a Venn diagram.
- How many students took part in the survey?

Let $\mathcal{E} = \{\text{all students that took part in the survey}\}$, $G = \{\text{those who like geography}\}$, $H = \{\text{those who like history}\}$, $M = \{\text{those who like mathematics}\}$.

- Fig. 8.5 is a Venn diagram containing the given information. The numbers in the regions represent the numbers of students in each region. (Check that the data in Fig. 8.5 agree with those in the question.)

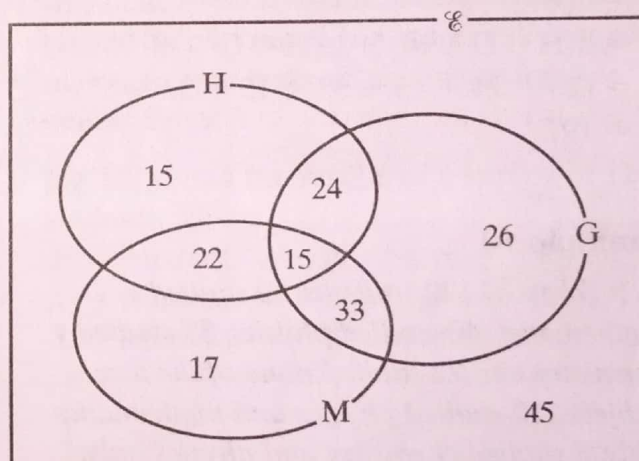


Fig. 8.5

Read the following carefully. It shows how the numbers in the regions of Fig. 8.5 were obtained.

$$\begin{aligned}\text{Since } n(H \cap G \cap M) &= 15 \text{ and} \\ n(H \cap G) &= 39 \\ n(H \cap G \cap M') &= 39 - 15 \\ &= 24\end{aligned}$$

Similarly

$$\begin{aligned}n(H \cap M \cap G') &= 37 - 15 \\ &= 22\end{aligned}$$

and

$$\begin{aligned}n(G \cap M \cap H') &= 48 - 15 \\ &= 33\end{aligned}$$

The number of students who like history only

$$\begin{aligned} &= n(H) - 15 - 22 - 24 \\ &= 76 - 61 \\ &= 15 \end{aligned}$$

Similarly, the number of students who like geography only

$$\begin{aligned} &= n(G) - 15 - 24 - 33 \\ &= 98 - 72 \\ &= 26 \end{aligned}$$

And the number of students who like mathematics only

$$\begin{aligned} &= n(M) - 15 - 33 - 22 \\ &= 87 - 70 \\ &= 17 \end{aligned}$$

- b** Number of students that took part in the survey
- $$\begin{aligned} &= n(\mathcal{U}) \\ &= n(H \cup G \cup M) + n(H \cup G \cup M)' \\ &= (26 + 24 + 15 + 33 + 15 + 22 + 17) + 45 \\ &= 197 \end{aligned}$$

Example 10

In a group of 120 students, 60 studied mathematics, 40 studied physics, 55 studied chemistry and 22 studied none of the three subjects. 12 studied physics and mathematics only, 8 studied chemistry and physics only, 7 studied mathematics and chemistry only.

- a** Draw a Venn diagram illustrating the information.
- b** Find the number of students that studied all three subjects.
- c** Find the number of students that studied only mathematics.

Let $\mathcal{U}$ = {all the students in the class},
 M = {those that studied mathematics},
 P = {those that studied physics},
 C = {those that studied chemistry}.

Let $n(M \cap P \cap C) = x$.

- a** Fig. 8.6 is a Venn diagram containing the given information. Check Fig. 8.6 carefully against the data.

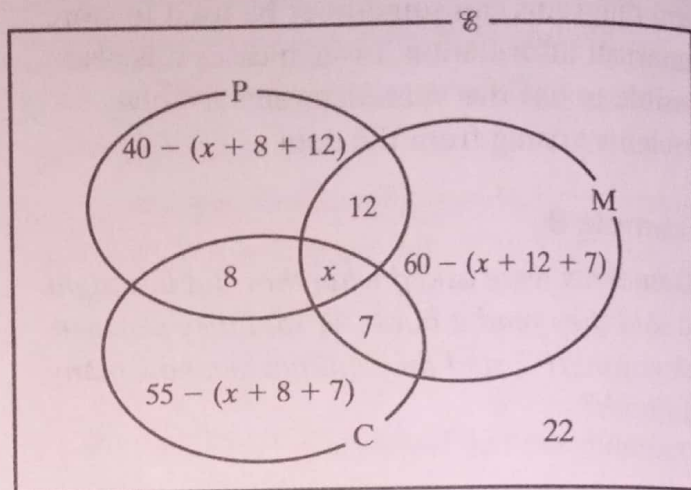


Fig. 8.6

Since 12 studied physics and mathematics only,

$$n(P \cap M \cap C') = 12.$$

Similarly, $n(C \cap P \cap M') = 8$

and $n(M \cap C \cap P') = 7$.

Again, since 60 studied mathematics, the number that studied mathematics only

$$\begin{aligned} &= n(M \cap P' \cap C') \\ &= n(M) - [n(M \cap P \cap C) \\ &\quad + n(M \cap P \cap C') \\ &\quad + n(M \cap C \cap P')] \\ &= 60 - (x + 12 + 7) \\ &= 60 - (x + 19) \end{aligned}$$

Similarly $n(P \cap M' \cap C') = 40 - (20 + x)$
 and $n(C \cap M' \cap P') = 55 - (15 + x)$.

- b** The total for the regions must add up to the number of students in the class.

$$\begin{aligned} &40 - (20 + x) + 60 - (19 + x) + 55 - (15 + x) + 12 + 8 + 7 + x + 22 = 120 \\ \Leftrightarrow &155 - (54 + 3x) + 49 + x = 120 \\ \Leftrightarrow &150 - 2x = 120 \\ \Leftrightarrow &x = 15 \end{aligned}$$

15 studied all three subjects.

- c** The number that studied only mathematics
- $$\begin{aligned} &= n(M \cap P' \cap C') \\ &= 60 - (19 + 15) \\ &= 26. \end{aligned}$$

In general:

- 1 Identify the sets, including the universal set.
- 2 Draw a Venn diagram.
- 3 Enter the data on the Venn diagram, starting with x , the unknown quantity.
- 4 Form an equation using the fact that the total number of elements in the regions equals the number of elements in the universal set.

Exercise 8d

Draw a suitable Venn diagram in each question.

- ① In Fig. 8.7 the numbers of elements in each region of the Venn diagram are as given.

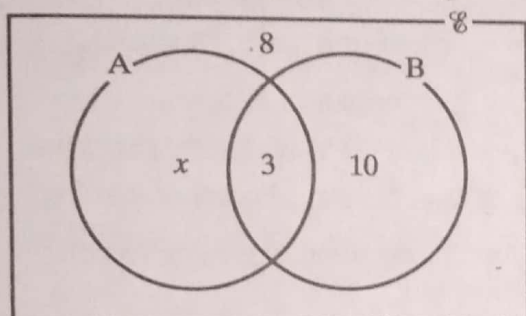


Fig. 8.7

If $n(U) = 30$ find

- a $n(A \cap B')$, b $n(A \cup B)$.

- ② In the Venn diagram in Fig. 8.8 the numbers of elements are as shown.

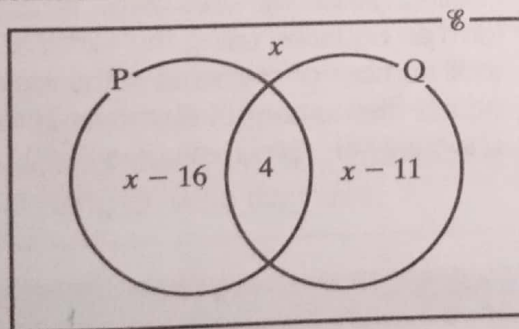


Fig. 8.8

Given that $n(P \cup Q) = n(P \cup Q)'$, find x and hence find $n(U)$.

- ③ A company employs 100 people, 65 of whom are men. 60 people, including all the women, are paid weekly. How many of the men are paid weekly?
- ④ In a village all the people speak Hausa or English or both. If 97% speak Hausa and 64% speak English, what percentage speak both languages?

- ⑤ 20 people apply for a job, 12 have school certificates and 10 have diplomas. If 2 have no qualifications, how many have both a school certificate and a diploma?
- ⑥ In a class of 35 students, 19 take history and 12 take economics. If 5 take both subjects, how many take neither?
- ⑦ In a school of 750 students, 320 are girls. 559 students do some kind of sport. If 101 girls do no sport, how many boys also do no sport?
- ⑧ In a survey of villagers, it is found that 20% of the people have visited Kano and 25% have visited Port Harcourt. If 5% have been to both cities, what percentage have visited neither place?
- ⑨ Out of 25 teachers, 16 are married and 15 are women. If 6 of the men are married, how many of the women are not married?
- ⑩ In a class of 36 students, 29 do mathematics and 20 do chemistry. If 5 students do neither, how many students do chemistry but not mathematics?
- ⑪ Fig. 8.9 shows the results of a survey of 126 students, where
 $F = \{\text{students who like playing football}\}$,
 $V = \{\text{students who like playing volleyball}\}$,
 $T = \{\text{students who like playing tennis}\}$.

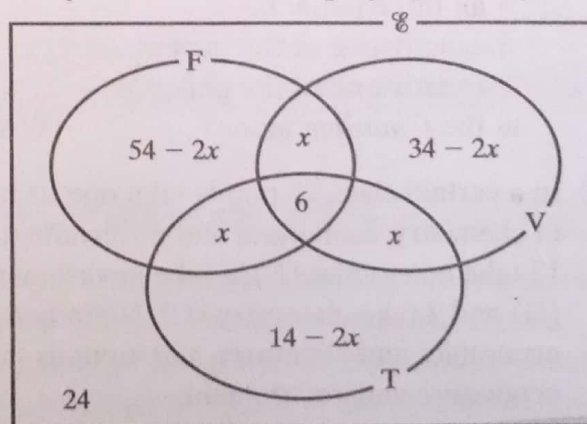


Fig. 8.9

- a How many students like playing none of the games?
- b How many students like playing football?
- c What is the value of x ?
- d How many students like playing tennis only?

- 12 A school sports team contains 68 students. 33 do field events, 40 do track events, 23 do swimming, 14 do both field and track events, 8 do both swimming and field events. If 15 students do field events only, and 10 do both swimming and track events, how many students do

- swimming only,
- track events only,
- all three events?

- 13 110 members of a sports club play at least one of the games, football, basketball and volleyball. If 20 play football and basketball only, 15 play football and volleyball only, 26 play basketball and volleyball only, x play all the three games, $2x$ each play only one game, how many play basketball altogether?

- 14 In a survey of 290 newspaper readers, 181 of them read the *Daily Times*, 142 read the *Guardian*, 117 read *Punch* and each reads at least one of the three papers. If 75 read the *Daily Times* and the *Guardian*, 60 read the *Daily Times* and *Punch*, and 54 read the *Guardian* and *Punch*:

- Draw a Venn diagram to illustrate this information.
- How many readers read
 - all three papers,
 - exactly two of the papers,
 - exactly one of the papers,
 - the *Guardian* alone?

[WAEC]

- 15 In a certain class, 22 pupils take one or more of chemistry, economics and government. 12 take economics (E), 8 take government (G) and 7 take chemistry (C). Nobody takes economics and chemistry, and 4 pupils take economics and government.

- Using set notation and the letters indicated above, write down the two statements in the last sentence.
- Draw a Venn diagram to illustrate the information.
- How many pupils take
 - both chemistry and government?
 - government only?

SUMMARY

- The *complement* of A , written A' , is the set that contains all those elements of $\mathcal{E}$ or $\mathcal{U}$ that are not members of A .
- If A and B are any two sets, then it is always true that $(A \cap B)' = A' \cup B'$ and $(A \cup B)' = A' \cap B'$.
- Sets of values can be written in set-builder notation, e.g.

$$R = \{x \in \mathbb{Z}: -1 < x < 5\}$$
 read as 'R is the set of integers, such that x lies between -1 and 5 '. Note that the letter $\mathbb{Z}$ = set of integers and that $-1 < x < 5$ is an example of a range of values.
- $\mathbb{Z}$ or $\mathbb{Z}$ is the set of integers.
 $\mathbb{N}$ or $\mathbb{N}$ is the set of natural numbers.
 $\mathbb{Q}$ or $\mathbb{Q}$ is the set of rational numbers.
- When using Venn diagrams to solve problems:
 - identify the sets, including the universal set;
 - decide on the unknown (i.e. what is required to be found);
 - draw a suitable Venn diagram;
 - enter the data in the Venn diagram, starting with x , the unknown;
 - form an equation using the fact that the total number of elements in the regions equals the number of elements in the universal set. (See Examples 8–10.)

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Identify and form open and closed simple statements
- 2 Deduce the truth or otherwise of simple statements
- 3 Form the negation of a simple statement
- 4 Distinguish between simple and compound statements
- 5 Draw conclusions from a given implication
- 6 Distinguish between conjunction and disjunction, representing them on truth tables
- 7 Recognise equivalent statements and apply them to arguments
- 8 Recognise and use the symbols for negation, conjunction, disjunction, implication and equivalence
- 9 Use Venn diagrams to demonstrate connections between statements.

Teaching and learning materials

Teacher: Wall charts showing truth tables; loops of string for Venn diagrams.

9.1 Simple statements

Introduction

In everyday life we use language in many different ways. We ask questions ('Where is the house?'); we give answers ('It is over there.');

we make statements ('I am a student.');

we make requests ('Please pass me that pen.');

we give commands ('Be quiet.');

we have sayings ('Nothing is permanent.') and so on. If an expression has a verb, it is a **sentence**. All of the previous examples are sentences. But we often communicate without verbs: 'What a lovely

picture!'; 'A tall student.'; 'Poetry in motion.'; '600 mm.' None of these has a verb; therefore, they are not sentences.

One way in which people use language is to try to develop a line of argument, or **reasoning**, that convinces other people. We often call this **logical reasoning**. In logical reasoning, a **statement** is a spoken or written sentence that gives information about something. A statement may be **true** or **false**.

Example 1

Decide whether or not each of the following is a sentence and/or a statement. Also say whether it is true or false.

- Lagos is a city in Nigeria.*
- Who is she?*
- They are lovely people.*
- $3 \times 2 = 5$*
- Submit your assignment.*
- God willing.*
- I saw them last week.*
- A square is a rectangle.*

Answers are given in Table 9.1 (overleaf), where Y stands for Yes and N stands for No.

Open and closed statements

In Table 9.1, some statements are either true or false. We call those **closed statements**. Closed statements are about well-defined things. Other statements, such as *c*) and *g*), are *not* about well-defined situations. We don't know if they are true or false. We call those **open statements**. In mathematics we use capital letters to show statements as in Example 2.

In a logical context, questions, commands, sayings and exclamations are not considered to be statements. This is why these are 'not applicable' answers in Example 2.